B. Serval and Inversion Magic

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, implementation, strings, two pointers |
https://codeforces.com/problemset/problem/1789/B

Serval has a string s that only consists of `0` and `1` of length n. The i-th character of s is denoted as s_i, where $1 \le i \le n$.

Serval can perform the following operation called Inversion Magic on the string s:

- Choose an segment $[l, r]$ ($1 \le l \le r \le n$). For $l \le i \le r$, change s_i into 1 if s_i is `0`, and change s_i into `0` if s_i is 1.

For example, let s be `010100` and the segment [2,5] is chosen. The string s will be `001010` after performing the Inversion Magic.

Serval wants to make s a palindrome after performing Inversion Magic **exactly once**. Help him to determine whether it is possible.

A string is a palindrome iff it reads the same backwards as forwards. For example, `010010` is a palindrome but `10111` is not.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains a single integer n ($2 \le n \le 10^5$) — the length of string s.

The second line of each test case contains a binary string s of length n. Only characters `0` and 1 can appear in s.

It's guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print `Yes` if s can be a palindrome after performing Inversion Magic exactly once, and print `No` if not.

You can output `Yes` and `No` in any case (for example, strings yEs, yes, Yes and YES will be recognized as a positive response).

Example

Input
3 4 1001 5 10010 7 0111011

Output
Yes Yes No

Note

In the first test case, Serval can perform Inversion Magic on the segment [1,4]. The string *s* will be `0110` after the magic.

In the second test case, Serval can perform Inversion Magic on the segment [1,3]. The string *s* will be `01110` after the magic.

In the third test case, Serval can't make *s* a palindrome by performing Inversion Magic exactly once.